

OpenWrt One/AP-24.XY: Revolutionising routers with Banana Pi

In a landmark partnership, OpenWrt developers are teaming up with Banana Pi to introduce the OpenWrt One/AP-24.XY router board.

The preliminary specifications of the OpenWrt One/AP-24.XY boast a MediaTek MT7981B (Filogic 820) dual-core Cortex-A53 processor, 1GB DDR4 system memory, and dual-band WiFi 6 via MediaTek MT7976C. The device is designed with affordability in mind, featuring a target price set below \$100.

Banana Pi's strategic selection as a partner stems from its boards closely aligning with OpenWrt's vision, popularity within the community, and existing support for essential features in upstream/mainline U-Boot and Linux. While certain components like the 2.5GbE PHY and Wi-Fi firmware remain non-open source, OpenWrt maintains transparency by making the schematics publicly available under an open source licence.

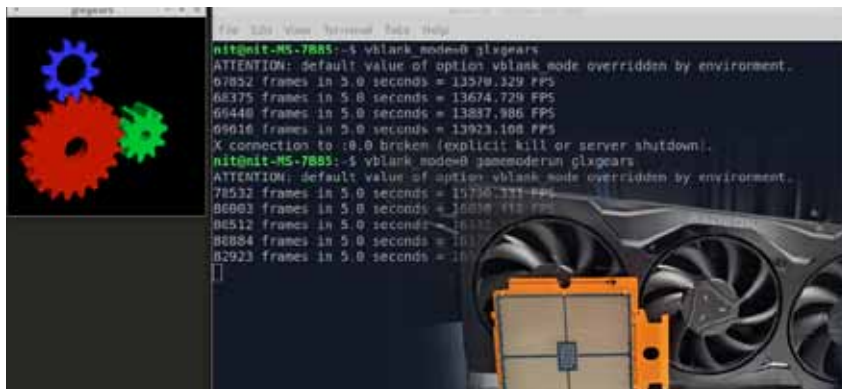
Beyond development, the collaboration introduces a revenue stream for OpenWrt. Banana Pi will handle manufacturing and distribution, pledging a share of the sales to the Software Freedom Conservancy (SDC) earmarked for OpenWrt. This financial model aims to cover hosting and conference expenses.

The OpenWrt One/AP-24.XY is poised for a 2024 launch, offering a glimpse into the future of routers.



contributions to the Mesa project. The recent update focuses on optimising the ACO compiler within the RADV driver, a Vulkan-based driver designed specifically for AMD GPUs running on Linux.

Mesa, originating as an open source driver for the OpenGL graphics API, has evolved into a collaborative effort involving developers from major tech players like AMD, Google, and Intel. RADV, an integral component of Mesa, utilises Valve's custom ACO compiler to streamline the conversion of shader programs into a format executable by the GPU.



The latest improvements result in a reduction of GPU instructions, enhancing code efficiency and subsequently boosting runtime performance. According to Phoronix, some users have experienced performance gains of up to 14% in games like Cyberpunk 2077, although specific circumstances for such increases remain unclear.

Valve's involvement in this initiative stems from the Linux-based SteamOS running on the Steam Deck. The AMD custom processor powering the handheld PC relies on the RADV driver, prompting Valve to refine the ACO compiler for better performance. Importantly, Valve's contributions to Mesa benefit not only the Steam Deck but also the broader Linux community, as the open source driver can be downloaded and utilised on any PC.

The collaborative nature of the Mesa project allows AMD's graphics developers to examine the changes made by Valve and potentially integrate them into their compiler. However, ACO's specialisation in RADV, as opposed to AMD's Linux drivers, may limit direct implementation.

For Steam Deck owners, the advice is to keep both software and drivers up-to-date to partake in the benefits of the improved compiler performance.

Intel releases update to its open source GPU compute stack

Intel has released CR 23.39.27427.23, an update to its open source GPU compute stack. Designed for Windows and Linux, this release caters to OpenCL and oneAPI



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